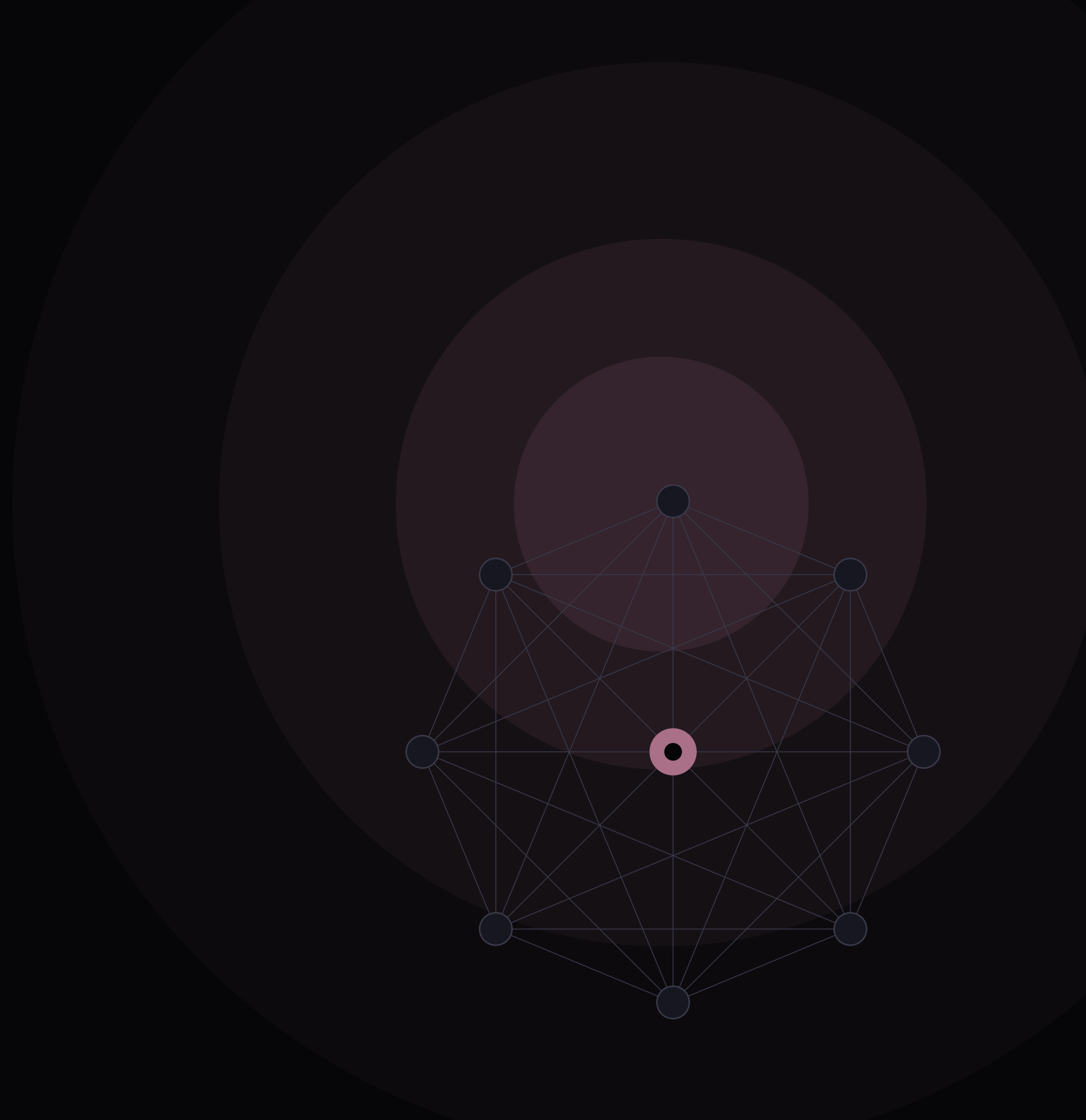


The model is the same. The system is the variable.

Roko is the open-source cognitive runtime. Korai is the agent-native coordination chain. Together they are the scaffold the next decade of production agents will compound against.

The model is commoditizing. The scaffold is the moat.

Models execute. Nunchi coordinates.



01 · EXECUTIVE THESIS

The model is commoditizing.

The scaffold is the moat.



MODEL

Executes

OpenAI · Anthropic · Google · OSS

RUNTIME

Coordinates

Roko · plan · gate · replay

CHAIN

Compounds

Korai · memory · reputation

PRIMITIVE

Settles

ISFR · clearing · yield perps

02 · NARRATIVE SPINE

Read the argument as a sequence, not a stack.

Five movements. One thesis. The PDF is structured to be skimmed top-to-bottom and read in any order.

01

PROBLEM

- Adoption stalls
- Coordination > capability
- Evidence: MAST
- Evidence: AIAM

02

CATEGORY

- Agent Coordination Plane
- What the plane owns
- Why retired frames fail

03

PRODUCT

- Roko runtime
- Korai chain
- Signal · Pulse · Cell
- Gate pipeline
- HDC memory

04

WEDGE & MOAT

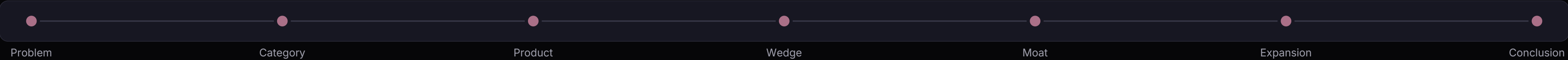
- Cost wedge stack
- Scaffold flywheel
- Knowledge compounding
- Network effects

05

EXPANSION

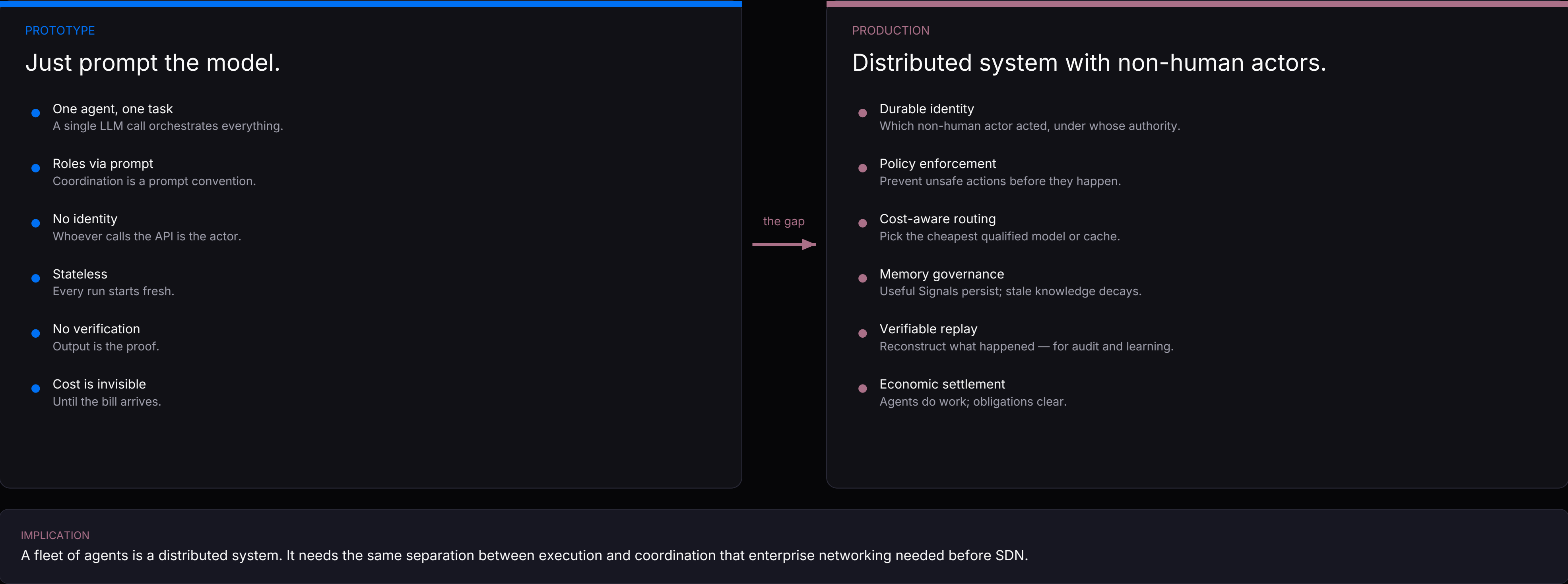
- ISFR benchmark
- Cooperative clearing
- Yield perpetuals
- Protocol economics

FLOW



03 · PROBLEM

Agents are crossing the prototype boundary. The missing layer is coordination.



04 · EVIDENCE

Failure is structural. The taxonomy is public.

Multi-agent systems fail in patterned ways. Cost discipline is under-weighted. Detection alone is insufficient.

TAXONOMY

1,600+ traces, 14 failure modes¹

MAST analyzes traces across seven multi-agent frameworks and groups failures into system design, inter-agent misalignment, and task verification.

κ = 0.88

annotator agreement

BENCHMARK DESIGN

Cost is under-weighted²

AI Agents That Matter argues evaluation overemphasizes accuracy and conflates model-developer with application-developer needs. Optimize cost and accuracy jointly.

joint

cost × accuracy

SECURITY ARCH

Architecture, not detection³

CaMeL extracts control/data flows so untrusted data cannot influence privileged paths. It solves 77% of AgentDojo tasks with provable security vs 84% undefended.

77 / 84

secure / undefended (AgentDojo)

ADAPTIVE ATTACKS

Detection-only defenses fail under adaptive prompt injection.⁴

One paper bypasses eight defenses with more than 50% attack success rate. The engineering takeaway is architectural isolation; the strategic takeaway is a coordination plane that owns policy.

SOURCES

1. MAST — Multi-Agent Failure Taxonomy — <https://arxiv.org/abs/2503.13657>

2. AI Agents That Matter — <https://arxiv.org/abs/2407.01502>

3. CaMeL — capability-based agent security — <https://arxiv.org/abs/2503.18813>

4. Adaptive Prompt Injection Attacks — <https://arxiv.org/abs/2503.00061>

Why “Agent Coordination Plane” is the correct frame.

Borrowed from SDN: separate the plane that decides from the layer that executes.

LAYER CAKE

Each layer solves one thing. The plane decides above all of them.

APPLICATIONS

LangChain · CrewAI · Mastra · Agents SDK

COORDINATION PLANE

Nunchi — identity · routing · memory · policy · settlement

NUNCHI

INTEROP

MCP (tools) · A2A (agents)

IDENTITY

ERC-8004 · Keycard-like primitives

PAYMENTS

x402 · Stripe/Tempo-like rails

EXECUTION

Models · Tools · Durable execution

RETIRED FRAMES

Why each adjacent label fails.

Trust layer for agents

Crowded and abstract. ‘Trust’ is an outcome, not an architecture.

Agent OS

Overclaimed. Agents need coordination, not a metaphor OS.

Stripe for agents

Payment rails converge separately (x402 et al.).

Identity layer

Identity is a primitive inside the plane, not the category.

Blockchain for agents

Strategically misleading near-term; buyer wants production agents to work.

POSITIONING

For teams moving agents from prototype to production, Nunchi is the Agent Coordination Plane that makes agents cheaper, safer, more accountable, and more useful over time — because it separates coordination from execution and turns every run into durable memory, policy, and reputation.

SOURCES

1. Coinbase x402 documentation — <https://docs.cdp.coinbase.com/x402/welcome>

2. ERC-8004 — Trustless Agents — <https://eips.ethereum.org/EIPS/eip-8004>

3. MCP joins Agentic AI Foundation — <https://blog.modelcontextprotocol.io/posts/2025-12-09-mcp-joins-agentic-ai-foundation/>

4. Linux Foundation A2A announcement — <https://www.linuxfoundation.org/press/a2a-protocol-surpasses-150-organizations-lands-in-major-cloud-platforms-and-sees-enterprise-produ...>

Roko · Korai · ISFR — three layers, one thesis.

Roko proves the runtime today. Korai compounds it across organizations. ISFR is the first financial primitive on top.

ROKO

Cognitive runtime

Open-source Rust runtime for production agents

- Plan · score · route · compose · act
- Verify · write · react
- Gate pipeline (11 gates / 7 rungs)
- CascadeRouter · NeuroStore · EpisodeLogger
- Control plane and replay

POSTURE

Internally demonstrated — external benchmark publication still required.

KORAI

Coordination chain

Agent-native sovereign EVM L1 — settlement & shared state

- HDC precompile (cheap similarity & binding)
- ERC-8004 identity, reputation, validation
- ZK-HDC behavioral fingerprints
- Job markets & cooperative clearing
- On-chain knowledge substrate

POSTURE

Specification & sequencing complete — implementation deferred until usage justifies.

ISFR

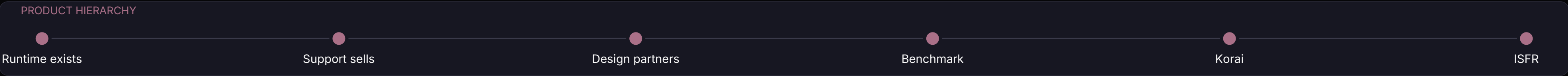
First financial primitive

Internet Secured Funding Rate · benchmark + perpetuals

- Validators publish benchmark inputs
- Track record before instruments settle
- Yield perpetuals create hedging activity
- Cooperative clearing (KKT-style)
- Reputation + knowledge as byproduct

POSTURE

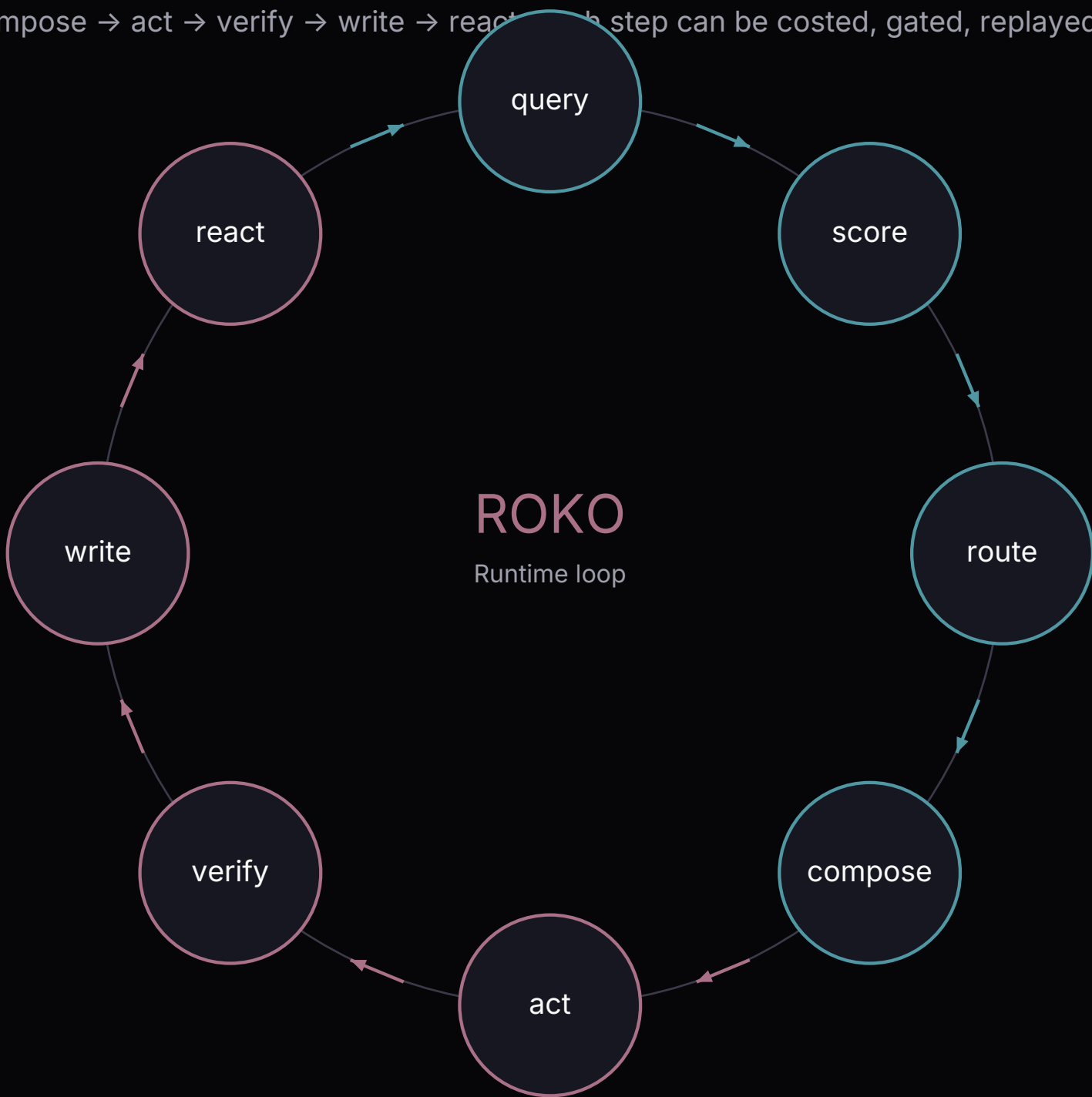
Expansion domain — proposed benchmark, not established market infrastructure.



07 · RUNTIME LOOP

Roko turns a model call into a governed event.

query → score → route → compose → act → verify → write → react → query. Each step can be costed, gated, replayed, and learned from.



WHY THE LOOP MATTERS

- Predict cost & risk**
Each step has a calibration target.
- Gate before commit**
Stop bad runs before they spend tokens.
- Persist usefully**
Signals are decay-aware; episodes are replayable.
- Learn from delta**
predict-publish-correct updates priors.
- Compose, don't hardcode**
'Wire, don't build' is the discipline.

STATE

Internally demonstrated: orchestration loop, gate pipeline (11/7), CascadeRouter, NeuroStore, EpisodeLogger, control plane. Roadmap: knowledge-informed routing default, dream consolidation scheduler, native harness as default.

08 · PRIMITIVES

Three primitives. One type system.

Signals are nouns, Pulses are events, Cells are computational verbs. Specializations are configured Cells, not new categories.

SIGNAL

(noun)

Durable, content-addressed knowledge

- Optional HDC fingerprint
- Decay metadata
- Replayable
- On-chain anchorable

PULSE

(event)

Ephemeral event on the bus

- Real-time
- Not persisted
- Bus-routed
- Triggers Cells

CELL

(verb)

Atomic computation unit

- Implements protocols
- Composable
- Stateless or stateful
- Configurable

9 PROTOCOLS PER CELL

A Cell can implement any combination — ‘wire, don’t build’.

Store

persist

Score

rank

Verify

check

Route

select

Compose

assemble

React

update

Observe

inspect

Connect

interop

Trigger

initiate

Coordination through traces, not just messages.

Agents leave Signals on a shared substrate; later agents read and refine. Like ants and pheromones — but typed, decayed, and verifiable.

MESSAGE-ONLY

Agents talk; coordination is ephemeral.

PROBLEMS

- No durable memory across sessions
- Failure cascades through chat history
- Replay = ‘read this transcript’

STIGMERGY

Agents modify a shared substrate.

PROPERTIES

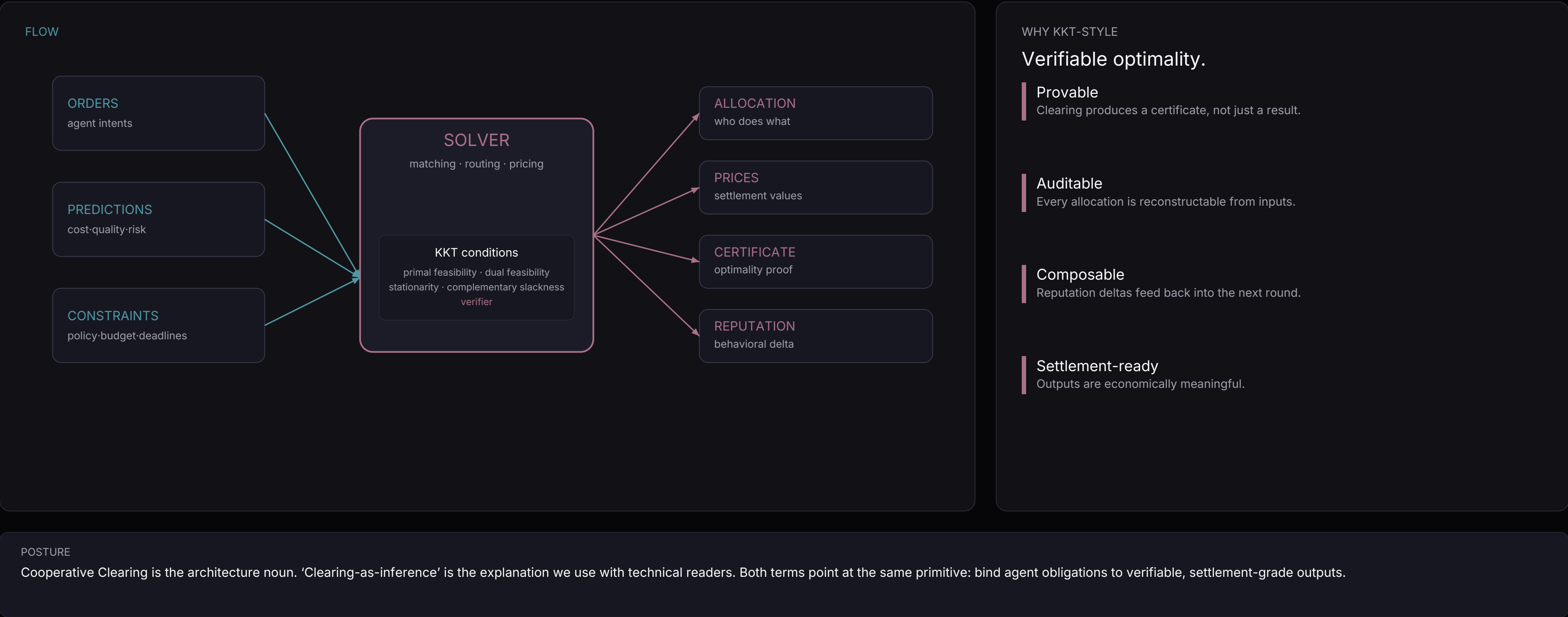
- Typed: Signals are content-addressed
- Decay: stale knowledge fades
- Verifiable: ZK-HDC fingerprints (roadmap)

WHY IT MATTERS

Stigmergy makes coordination compounding. The thousandth agent joins smarter than the first because it inherits the trace, not the transcript.

Clearing as inference. Verification as KKT-style optimality.

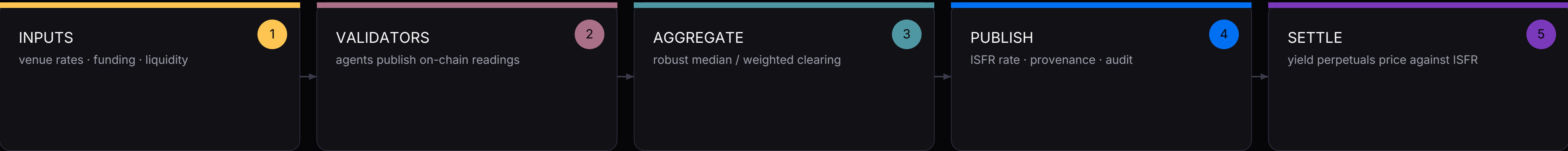
Match, route, settle, score — and prove the result is optimal under the published constraints.



11 · ISFR

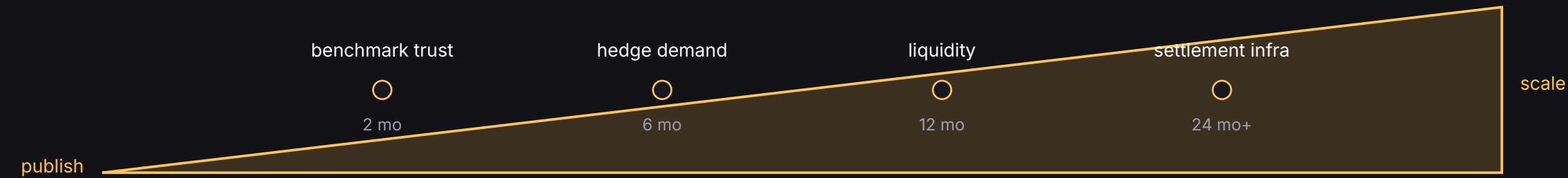
ISFR — Internet Secured Funding Rate.

A benchmark earns trust by being repeatedly produced before instruments settle against it. ISFR is the first financial primitive on the coordination network.



YIELD PERPETUAL WEDGE

Hedging activity creates real liquidity around the benchmark.



POSTURE

Proposed benchmark, not established market infrastructure. Frame ISFR as expansion domain after the coordination plane has earned trust. Do not lead investor materials with ISFR unless audience is crypto-native or financial-infrastructure-focused.

ANCHOR

BIS reports USD 7.9 T avg daily OTC interest-rate-derivatives turnover (Apr 2025), incl. USD 5.1 T OIS. ¹

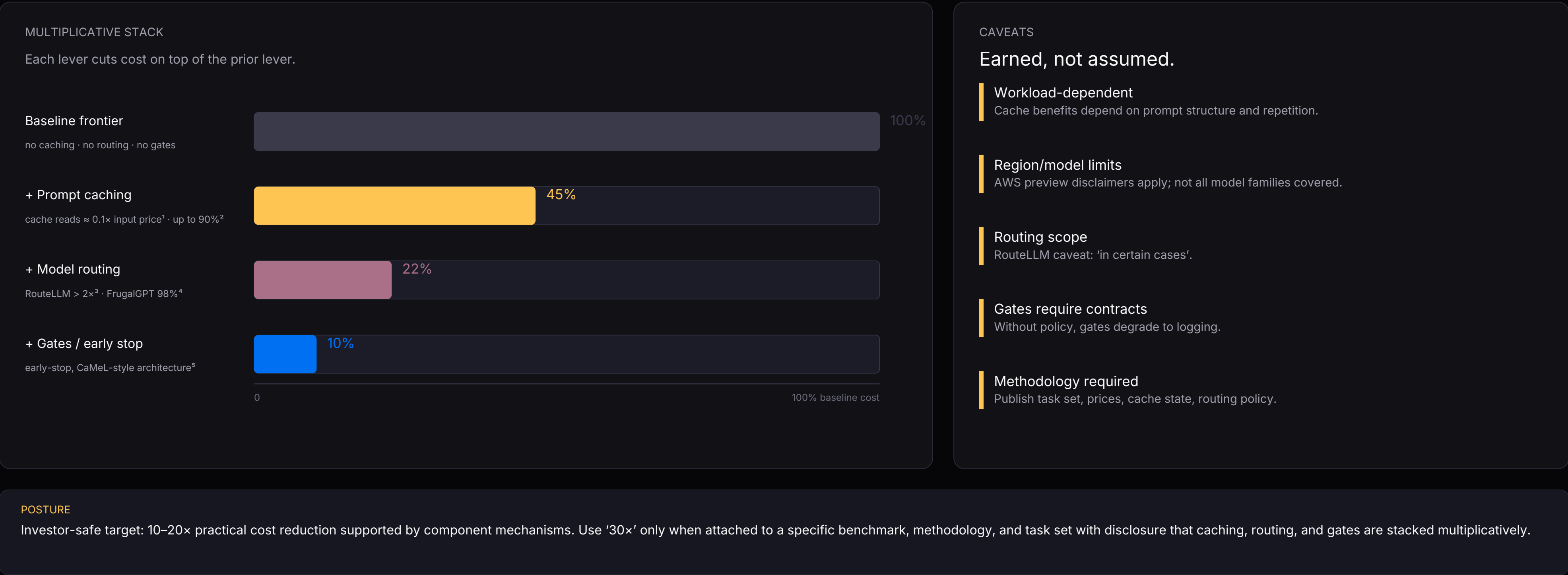
Use to motivate scale of benchmark-rate infrastructure — not to imply ISFR captures that TAM.

SOURCES

1. BIS OTC IRD turnover, April 2025 — https://www.bis.org/statistics/rpfx25_ir.htm

The wedge is multiplicative. Stack the mechanisms.

Caching, routing, and early-stop gates each have public evidence. Compounded, they make 10–20x practical cost reduction credible.



SOURCES

1. Claude prompt caching documentation — <https://docs.anthropic.com/en/docs/build-with-claude/prompt-caching>

2. AWS Bedrock prompt caching and routing — <https://aws.amazon.com/blogs/aws/reduce-costs-and-latency-with-amazon-bedrock-intelligent-prompt-routing-and-prompt-caching-pr...>

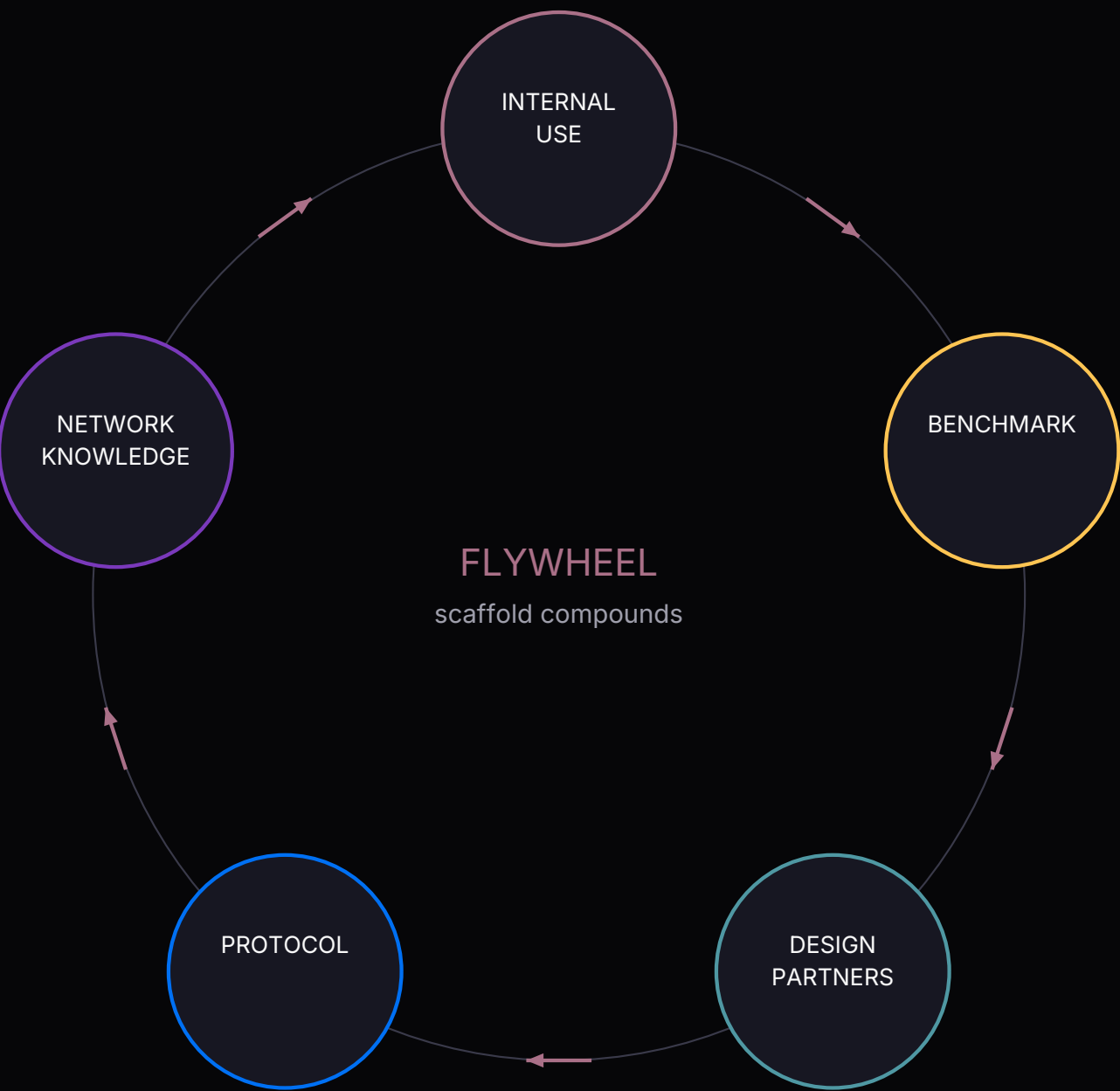
3. RouteLLM — <https://arxiv.org/abs/2406.18665>

4. FrugalGPT — <https://arxiv.org/abs/2305.05176>

5. CaMeL — capability-based agent security — <https://arxiv.org/abs/2503.18813>

Cost is the wedge. Knowledge is the moat.

Internal product use → benchmark → design partners → protocol surface → network knowledge. Each turn deepens the next.



FIVE-TURN SEQUENCE

Why each turn deepens the next.

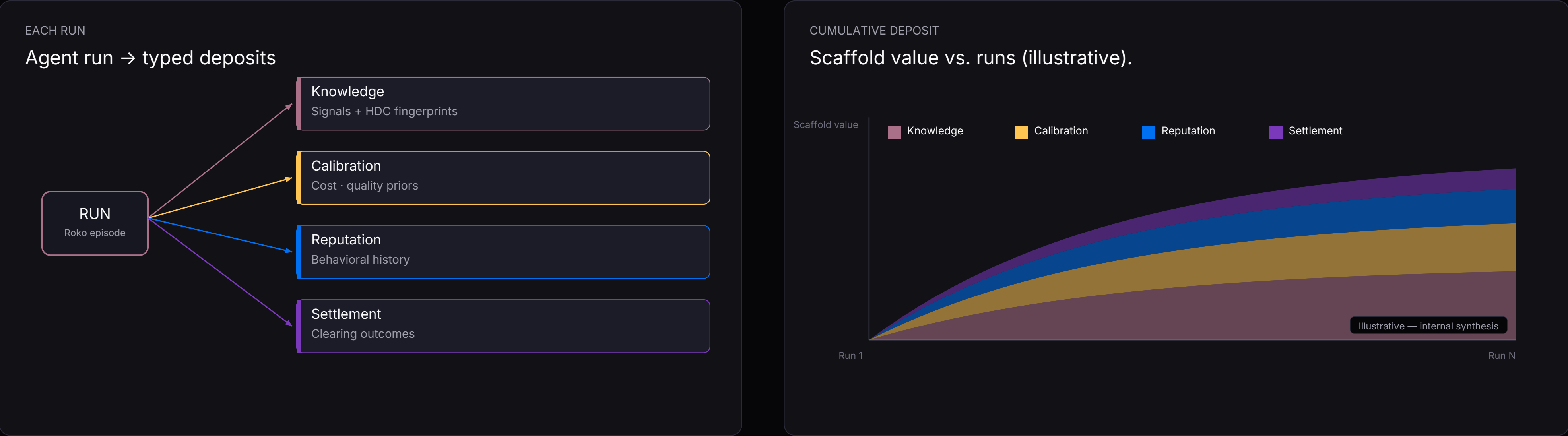
- 01 Internal use**
Roko inside the company first.
- 02 Benchmark**
Reproducible cost-quality methodology.
- 03 Design partners**
Real workloads from named partners.
- 04 Protocol surface**
MCP, A2A, ERC-8004 integrations.
- 05 Network knowledge**
Memory, calibration, reputation compound.

FRAME

'The thousandth agent joins smarter than the first.' Architectural goal and network-effect thesis — not a current-state metric. Inside one Roko deployment, local memory compounds today.

Every run leaves a deposit. The deposit is the moat.

Each agent run deposits durable artifacts into four reservoirs. New runs draw from them. Forks start empty.

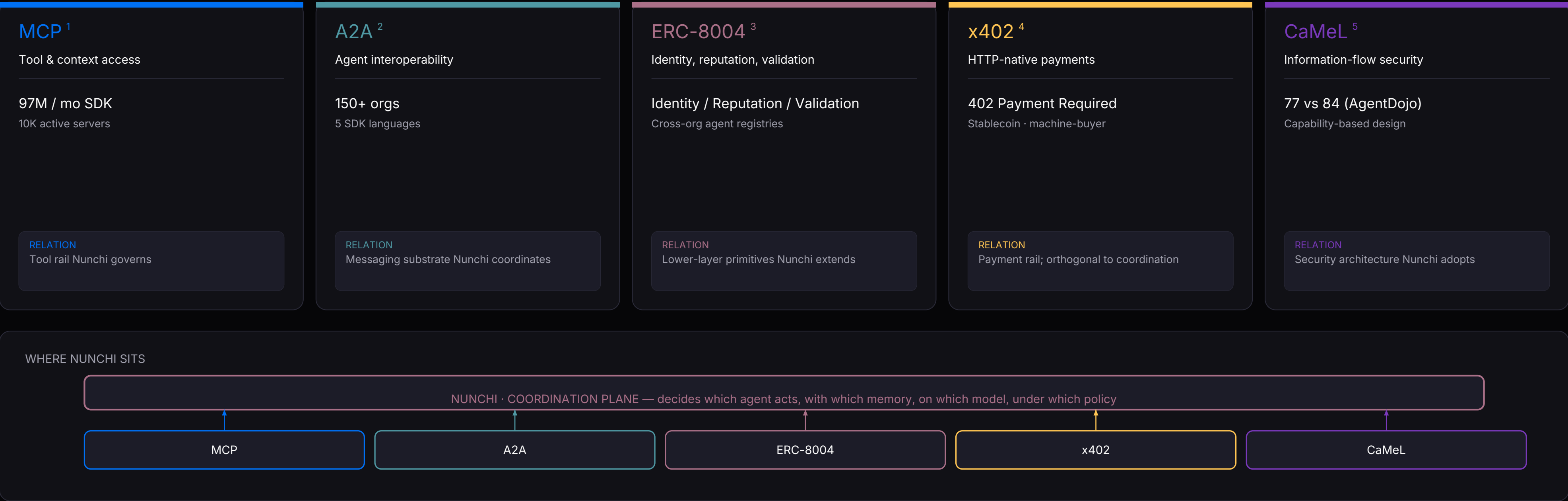


WHY IT MATTERS

Code can be forked. Lived coordination history cannot. The scaffold is the product because forks start empty — and the scaffold is what the buyer keeps even when models change underneath.

The protocols are crystallizing. The plane above them is missing.

MCP, A2A, ERC-8004, x402 each solve one rail. Nunchi sits above all of them.



SOURCES

1. MCP joins Agentic AI Foundation — <https://blog.modelcontextprotocol.io/posts/2025-12-09-mcp-joins-agentic-ai-foundation/>

2. Linux Foundation A2A announcement — <https://www.linuxfoundation.org/press/a2a-protocol-surpasses-150-organizations-lands-in-major-cloud-platforms-and-sees-enterprise-produu...>

3. ERC-8004 — Trustless Agents — <https://eips.ethereum.org/EIPS/eip-8004>

4. Coinbase x402 documentation — <https://docs.cdp.coinbase.com/x402/welcome>

5. CaMeL — capability-based agent security — <https://arxiv.org/abs/2503.18813>

HDC is a technical bet — not the whole bet.

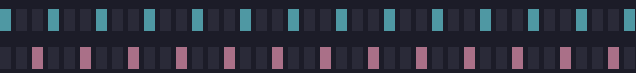
Hyperdimensional computing gives the system a cheap, composable, proof-friendly substrate. Production claims still need validation.

OPERATIONS

Bind · Bundle · Similarity · Cleanup

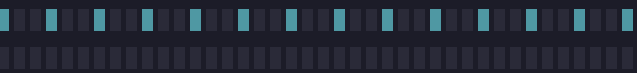
BIND

permute / multiply



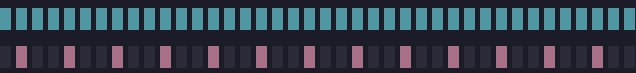
BUNDLE

majority / sum



SIMILARITY

Hamming distance



CLEANUP

associative recall



EVIDENCE

Promising. Not a market consensus.

PathHD¹

Single-LLM-call HDC for KG reasoning. 40–60% lower latency, 3–5× lower GPU memory vs neural baselines.

Bit-Level Semantics²

99% dense retrieval quality, 8× memory reduction, 860+ QPS at 10M docs, 7.68× higher throughput.

Commercial traction

Thin in the source corpus. Treat HDC as swappable except where binary, deterministic, proof-friendly properties matter.

POSTURE

HDC is attractive because it is cheap, deterministic, composable, and proof-friendly. It should be a Nunchi advantage — not a claim that the market has standardized on HDC.

SOURCES

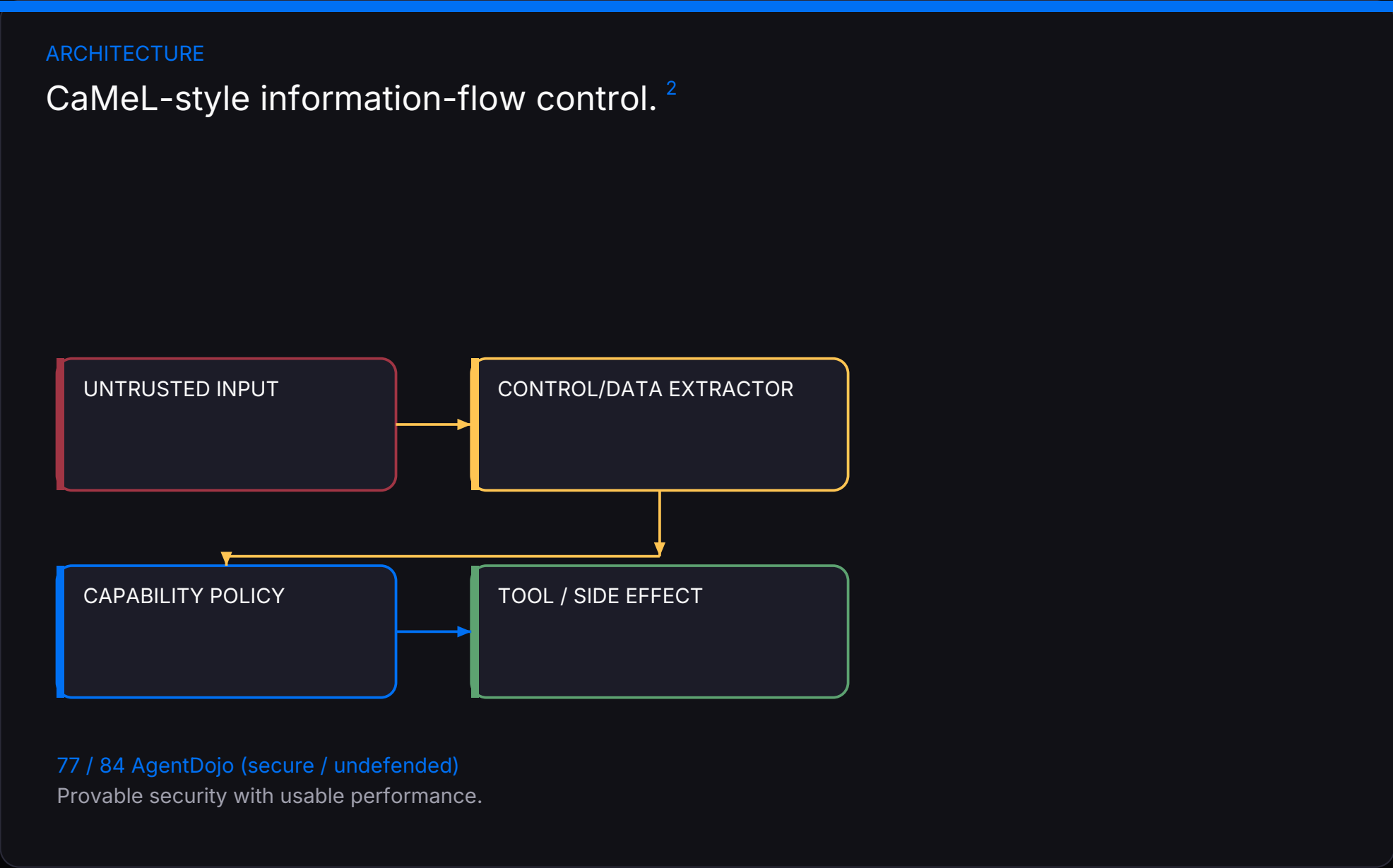
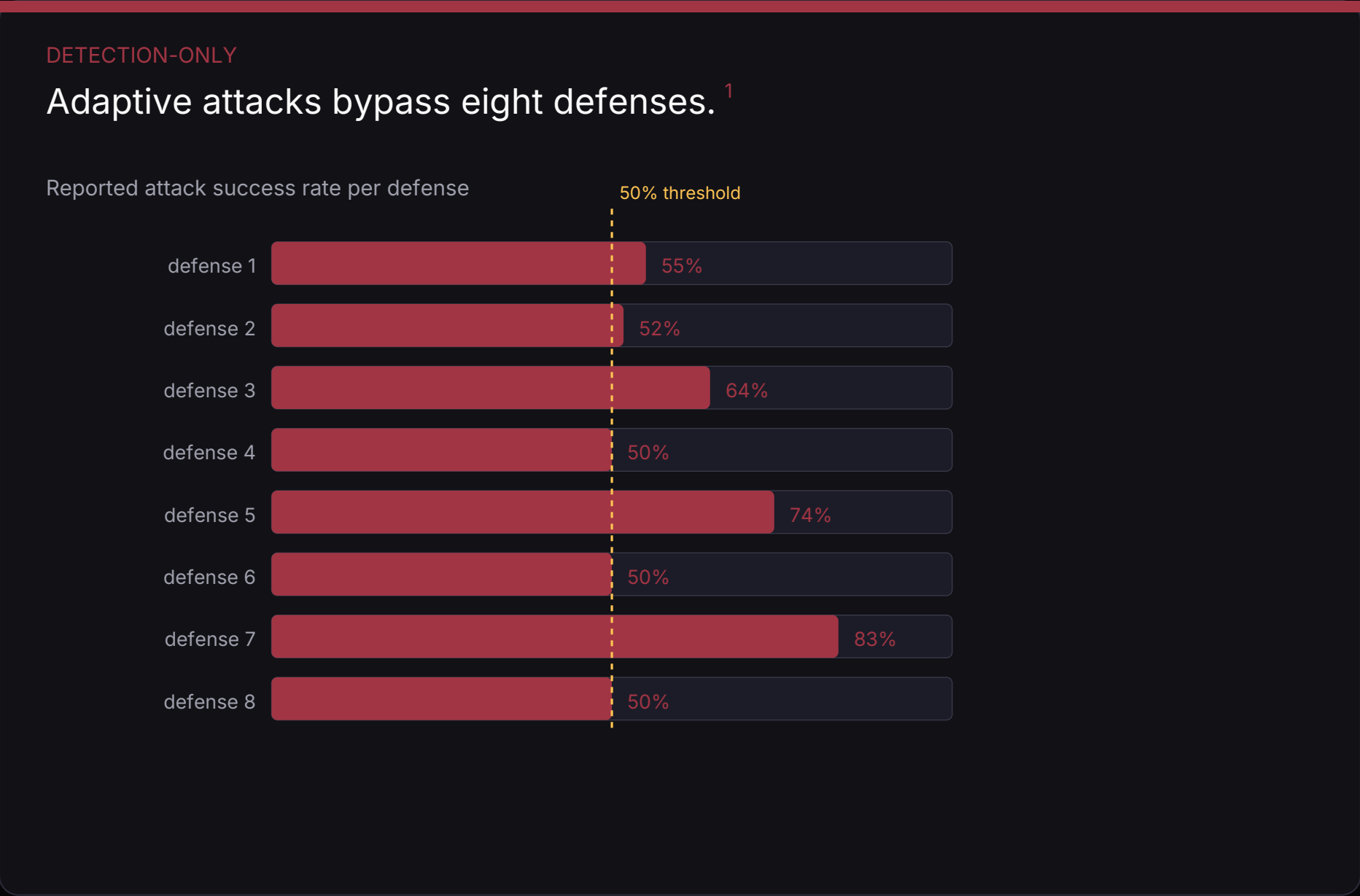
1. PathHD — encoder-free HDC for KG reasoning — <https://www.semanticscholar.org/paper/9ccbef593ba1e4eb5009b7477493d687f72bb9ed>

2. Bit-Level Semantics — HDC retrieval — <https://ieeexplore.ieee.org/document/11282933/>

17 · SECURITY ARCHITECTURE

CaMeL-style isolation > prompt-layer detection.

Untrusted data must not steer privileged paths. Capabilities, not heuristics, enforce tool policies.



POSITION

Treat information-flow control and sandboxing as architecture, not as add-ons. Detection-based defenses are useful as a layer, but CaMeL-style design is the right reference pattern for production agents that touch untrusted data.

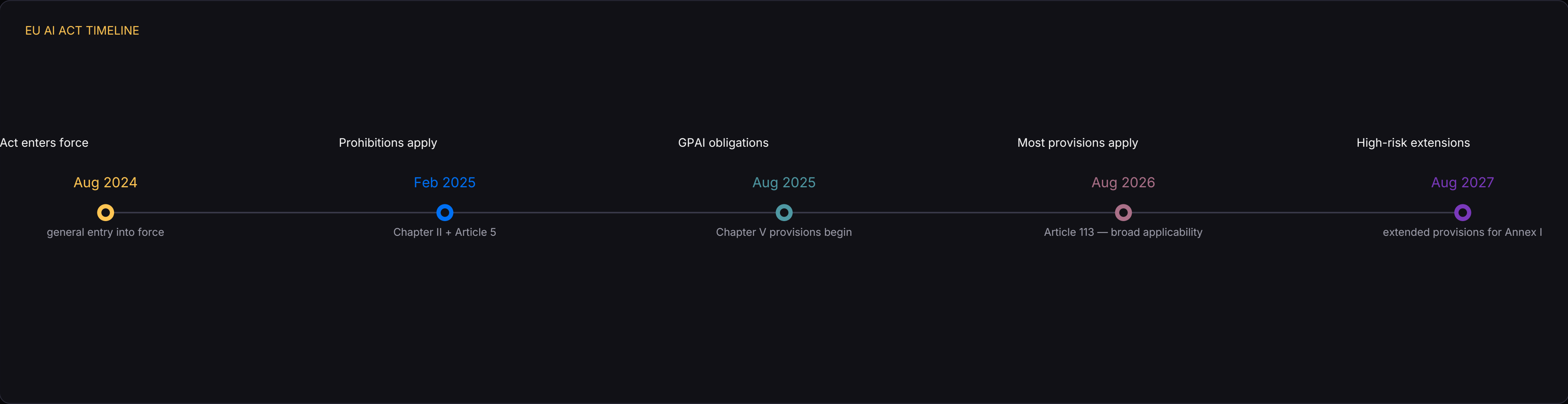
SOURCES

1. Adaptive Prompt Injection Attacks — <https://arxiv.org/abs/2503.00061>

2. CaMeL — capability-based agent security — <https://arxiv.org/abs/2503.18813>

Compliance is a byproduct of coordination.

Article 50 transparency. Article 99 penalties. Article 113 timing. Nunchi supports the artifacts; legal compliance is broader.



ARTICLE 50¹

Transparency obligations

Inform users · machine-readable marks · disclose deepfakes & public-interest AI text

Nunchi: provenance, identity, replay artifacts.

ARTICLE 99²

Penalties

Up to EUR 15M or 3% of worldwide turnover for Article 50 transparency failures (SME-adjusted)

Use carefully — not every customer faces maximum penalties.

ARTICLE 113³

Timing

Generally applies from 2 August 2026, with specified exceptions earlier and later

Sales urgency point — without overclaiming buyer readiness.

SOURCES

1. EU AI Act, Article 50 — <https://artificialintelligenceact.eu/article/50/>

2. EU AI Act, Article 99 — <https://artificialintelligenceact.eu/article/99/>

3. EU AI Act, Article 113 — <https://artificialintelligenceact.eu/article/113/>

Benchmark rates become infrastructure when they are trusted.

BIS data shows the eventual scale of rate-derivatives infrastructure. ISFR is not a TAM claim — it’s a primitive that fits if it earns adoption.

MARKET FACT

USD 846 T¹

Total OTC derivatives notional, end-Jun 2025

BIS — full OTC derivatives market

MARKET FACT

USD 668 T¹

Implied IRD notional (79% of OTC notional)¹

Derived from BIS 79% IRD share

MARKET FACT

USD 7.9 T²

Average daily OTC IRD turnover, Apr 2025

BIS — including USD 5.1 T OIS

INTEREST-RATE DERIVATIVES = 79% OF OTC NOTIONAL

Why a trusted benchmark matters at any scale.

IRD share — 79%

Other 21%

0%100%

POSTURE

Use these numbers to motivate the importance of benchmark-rate infrastructure — not to imply ISFR captures that traditional TAM. ‘Implied IRD notional’ is a derived calculation from BIS figures, not a separately published BIS line item.

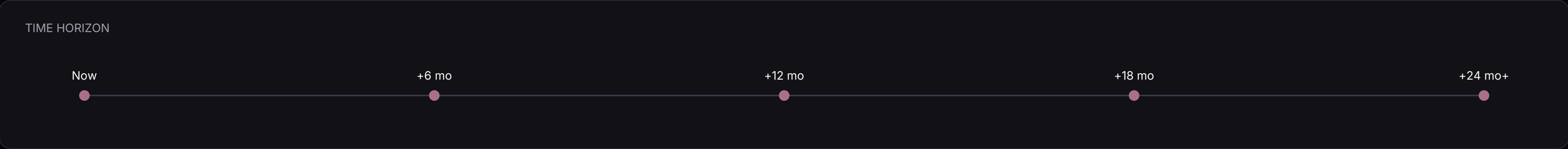
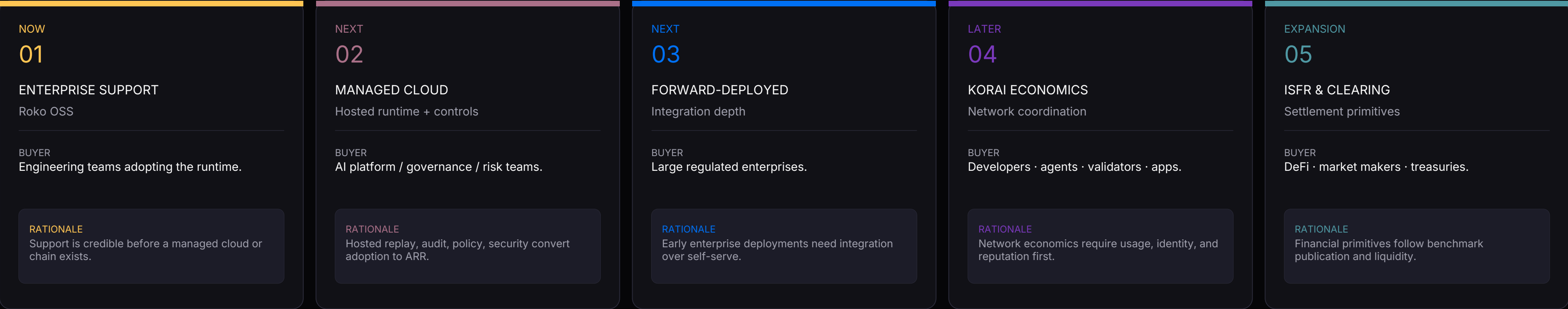
SOURCES

1. BIS OTC derivatives statistics, Jun-2025 — https://www.bis.org/publ/otc_hy2512.htm

2. BIS OTC IRD turnover, April 2025 — https://www.bis.org/statistics/rpfx25_ir.htm

Sell support before tokens. Sell tokens never, until usage exists.

Services revenue → enterprise runtime → coordination/benchmark primitives → protocol-native economics.

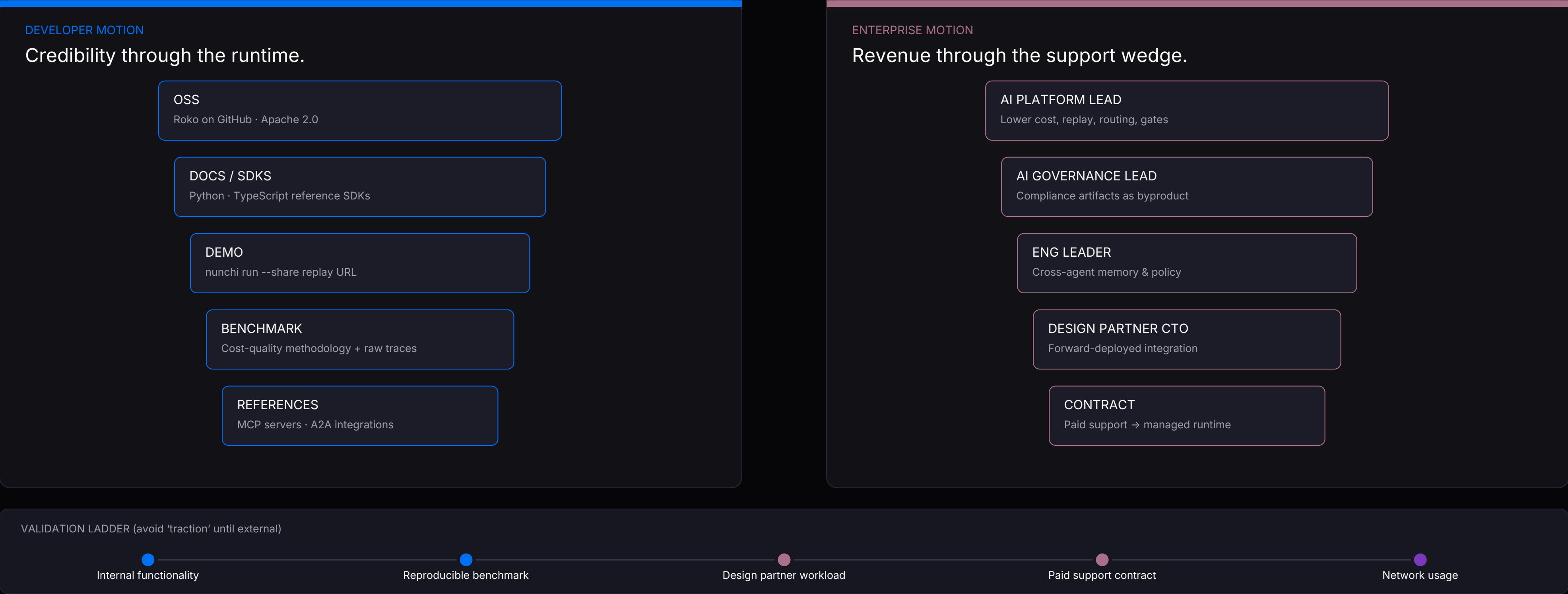


DISCIPLINE

Open-source Roko is distribution, not charity. Token language stays out of the investor-facing core narrative unless asked. The token path can be real without being the pitch.

Two motions, one funnel.

Developer adoption proves the runtime. Enterprise support converts adoption into revenue.



Sequencing is the strategy.

Roko first, Korai after proof, ISFR after Korai earns trust. Avoid building Korai before it's justified by usage.

NOW

0–6 months

Runtime hardening (no permissive fallbacks)

SDKs (Python · TypeScript) + reference MCP servers

Reproducible cost-quality benchmark + methodology

SOC 2 process, Article 50 mapping doc

Shareable demo: `nunchi run --share` replay URL

NEXT

6–12 months

Knowledge-informed routing (NeuroStore → CascadeRouter)

CaMeL-style information-flow control

Cross-org sync (permitted Signal exchange)

Load tests at larger agent counts

Named design partners + workloads

LATER

12–24 months

Korai testnet (sovereign EVM L1 + HDC precompile)

ERC-8004 identity / reputation / validation flows

ZK-HDC proofs for similarity & fingerprints

Job market on Korai (ERC-8183 style)

ISFR benchmark publication

LATER+

24+ months

Yield perpetuals & cooperative clearing live

Token-path economics (only after marketplace data)

Foundation/protocol structure if needed

Cross-chain settlement integrations

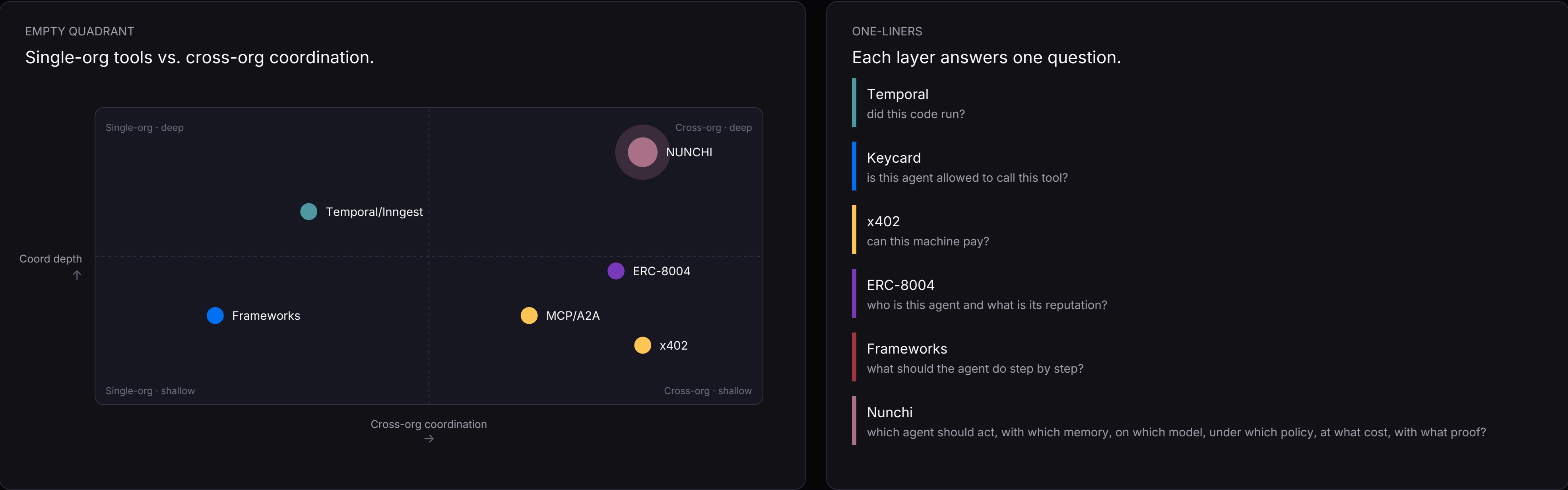
Global benchmark adoption

CRITICAL PATH

Developer adoption → reproducible benchmark → design-partner usage → enterprise support contract → Korai testnet (only after Roko has earned the chain) → ISFR publication (only after Korai has a track record).

Don't fight neighbors. Own the empty quadrant.

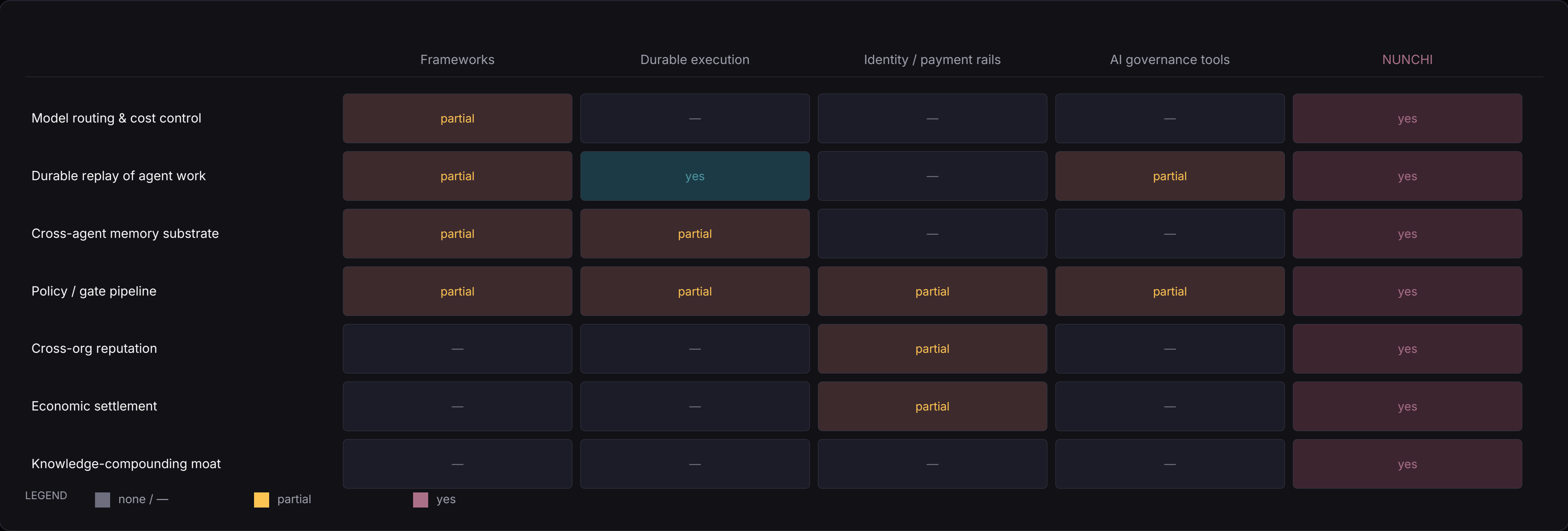
Most adjacent companies solve a rail or layer. Nunchi owns the decision plane above the rails.



POSITION
Nunchi is below frameworks and above rails. Owns cross-organization memory, identity, reputation, and settlement — exactly the gap that no incumbent owns.

Where Nunchi differs — by capability, not by vendor.

Each row is a capability. Each column is a class of competitor. Cell color = strength.

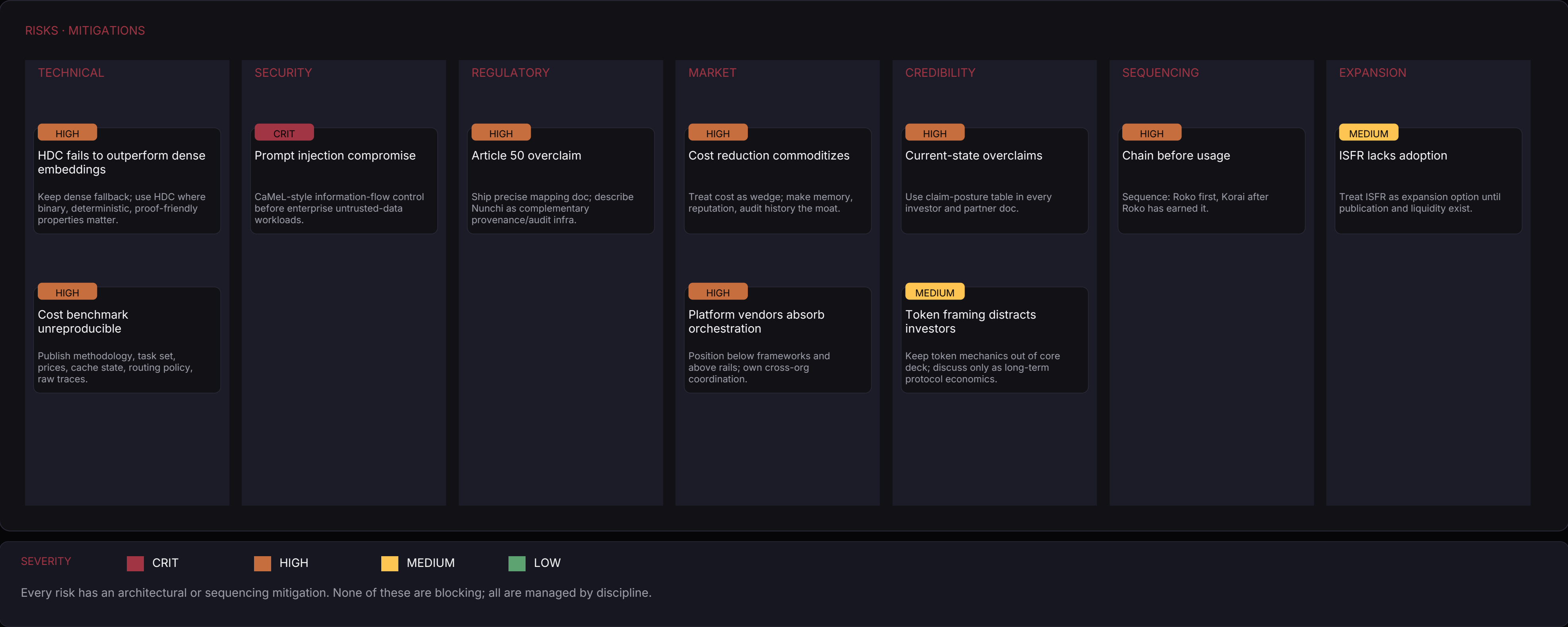


READING

Nunchi’s row of ‘yes’ does not claim those capabilities are fully shipped today. Each is mapped to product state — see roadmap and posture pages.

Risks are explicit. Mitigations are architectural.

Five risk classes. Severity by color. Mitigation tied to product or sequencing discipline.



What is real, what is internal, what is target, what is deferred.

Every claim is labeled. The discipline is not to over-promote — and to make the labels easy to find.

EXTERNALLY VALIDATED	INTERNALLY DEMONSTRATED	DESIGN TARGET	DEFERRED / SPECULATIVE
supported by published research or vendor docs	exists in source corpus / internal benchmarks	intended; methodology required to claim	not active; sequencing-bound
MAST taxonomy of multi-agent failures across 1,600+ traces 1 Adaptive attacks bypass eight prompt-injection defenses 2 CaMeL solves 77% of AgentDojo with provable security 3 Anthropic cache reads $\approx 0.1\times$ input pricing; AWS up to 90% 4 5 BIS USD 7.9T daily OTC IRD turnover (Apr 2025) 6	Roko orchestration loop wired internally Gate pipeline: 11 gates across 7-rung adaptive pipeline CascadeRouter with model-routing persistence EpisodeLogger and replay control-plane routes	10–20$\times$ practical cost reduction (with methodology) Cross-org Signal exchange between Roko deployments ZK-HDC behavioral fingerprints ISFR as a published benchmark with track record	Korai production chain live Agent job marketplace (ERC-8183 style) operational NUNCHI token launch Yield perpetuals trading on settled liquidity

SOURCES

1. MAST — Multi-Agent Failure Taxonomy — <https://arxiv.org/abs/2503.13657>

2. Adaptive Prompt Injection Attacks — <https://arxiv.org/abs/2503.00061>

3. CaMeL — capability-based agent security — <https://arxiv.org/abs/2503.18813>

4. Claude prompt caching documentation — <https://docs.anthropic.com/en/docs/build-with-claude/prompt-caching>

5. AWS Bedrock prompt caching and routing — <https://aws.amazon.com/blogs/aws/reduce-costs-and-latency-with-amazon-bedrock-intelligent-prompt-routing-and-prompt-caching-pre...>

6. BIS OTC IRD turnover, April 2025 — https://www.bis.org/statistics/rpfx25_ir.htm

Five minutes. Four primitives. One replay URL.

The demo shows identity, prediction, shared knowledge, and durability — and ends with a link the prospect leaves with.

01

R15 · run

```
$ nunchi run agent.toml --task R15
```

SHOWS

Identity · policy · allowed actions before execution.

PROVES
Identity

02

prediction

```
$ system predicts cost / risk / route
```

SHOWS

Predicted vs actual after run.
predict-publish-correct loop.

PROVES
Prediction

03

shared memory

```
$ second agent reuses Signals from first
```

SHOWS

Compounding scaffold across runs.

PROVES
Shared knowledge

04

kill / resume

```
$ interrupt + resume from snapshot
```

SHOWS

Production-grade durability. Replay is not a transcript.

PROVES
Durability

05

nunchi audit

```
$ nunchi audit + nunchi run --share
```

SHOWS

Inspectable trace. Replay URL leaves the room.

PROVES
Trust artifact

DISCLOSURE POSTURE

The best demo is a controlled system, not a magical agent.

Local simulation

Use until Korai is live. Say so.

Cached responses

Pre-warm explicitly when speed is deterministic.

ZK proofs

Distinguish local generation from on-chain verification.

Cost numbers

Tied to exact task set, prices, cache state, routing policy.

run --share

Treat as not-yet-built unless implementation has changed.

GOAL

End with a replay URL the prospect leaves with. Inspectable, interruptible, accountable. The takeaway is not the agent; it's the scaffold around it.

Three audiences. One thesis. Three levels of disclosure.

Same architecture, different framing. Public is the category creator; investor adds detail; internal is the unvarnished build plan.

PUBLIC WHITEPAPER	INVESTOR APPENDIX	INTERNAL ONLY
✓ Category thesis ✓ Roko architecture (summary) ✓ Korai chain (sequenced) ✓ ISFR (brief expansion) ✓ Risk register (summarized) ✓ Token design (minimal) — Detailed risk register — Design partner target list — a16z-specific pitch notes ✓ Implementation gaps (high level only)	✓ Category thesis ✓ Roko architecture (detailed) ✓ Korai chain (detailed) ✓ ISFR (detailed appendix) ✓ Risk register (full) ✓ Token design (optional, crypto-native) ✓ Implementation gaps (detailed) ~ Design partner shortlist — a16z-specific pitch notes ✓ Claim posture matrix	✓ Category thesis ✓ Roko architecture (full) ✓ Korai chain (full) ✓ ISFR (full) ✓ Risk register (full) ✓ Token design (full) ✓ Implementation gaps (full) ✓ Design partner target list ✓ a16z-specific pitch notes ✓ Build sequence + dependencies

PRINCIPLE

Investor-safe but not dumbed down. The same architecture flows through every disclosure level — only depth and audience-specific notes change.

Models execute. Nunchi coordinates.

THE FIVE-LINE VERSION

- 1. Agents create coordination failures that models alone do not solve.
- 2. Roko is the runtime wedge — production agents become cheaper, safer, replayable.
- 3. Korai turns local traces into cross-organization memory, reputation, validation, settlement.
- 4. ISFR demonstrates the first financial primitive a coordination network supports.
- 5. The company wins by owning the scaffold, not by owning a model.

TONE

Sell inevitability through architecture.

Not hype. Not aspiration. The category is correct because the systems problem is real, the protocols are crystallizing, and the scaffold compounds in ways code cannot.

The model is the same. The system is the variable.

The model is commoditizing. The scaffold is the moat.

preserve verbatim across materials

SHORT VERSION

Models execute. Nunchi coordinates.

WHAT TO BUILD

A disciplined runtime first. A reproducible benchmark next. Design partners after that. The chain only when usage justifies it. The financial primitive only after the chain has earned trust.

One word, one canonical meaning.

Use this column when writing or editing any Nunchi document.

Nunchi Company and umbrella coordination system. Use as the umbrella.	HDC Hyperdimensional / vector-symbolic computing. Cheap, deterministic, proof-friendly substrate.
Roko Open-source cognitive agent runtime. Current beachhead and first support product.	Predict-publish-correct Online learning loop: predict → publish → observe → emit delta → update.
Korai Agent-native coordination chain. Earlier docs sometimes say ‘Nunchi chain’ — standardize on Korai.	Gate pipeline Bridge between agent execution and enterprise acceptability. 11 gates, 7 rungs.
Agent Coordination Plane Category. Use instead of ‘trust layer’, ‘agent OS’, or ‘Stripe for agents’.	ISFR Internet Secured Funding Rate. Future expansion benchmark — not beachhead.
Cooperative Clearing Architecture pattern for matching, routing, settling, scoring agent obligations.	Trust Credits Internal accounting concept. Not a general-purpose payment promise.
Signal Durable, content-addressed memory object. The only noun in the Roko type system.	Token-path economics Use instead of ‘tokenomics’ to avoid crypto-first framing.
Pulse Ephemeral bus event. Real-time coordination event.	Compliance artifacts Provenance, identity, replay outputs. Not a ‘compliance solution’.
Cell Atomic computation unit. Replaces deprecated ‘Block’ terminology.	

Sentences to preserve. Phrases to retire.

Editorial discipline keeps the architecture story coherent across decks, memos, landing pages, and demos.

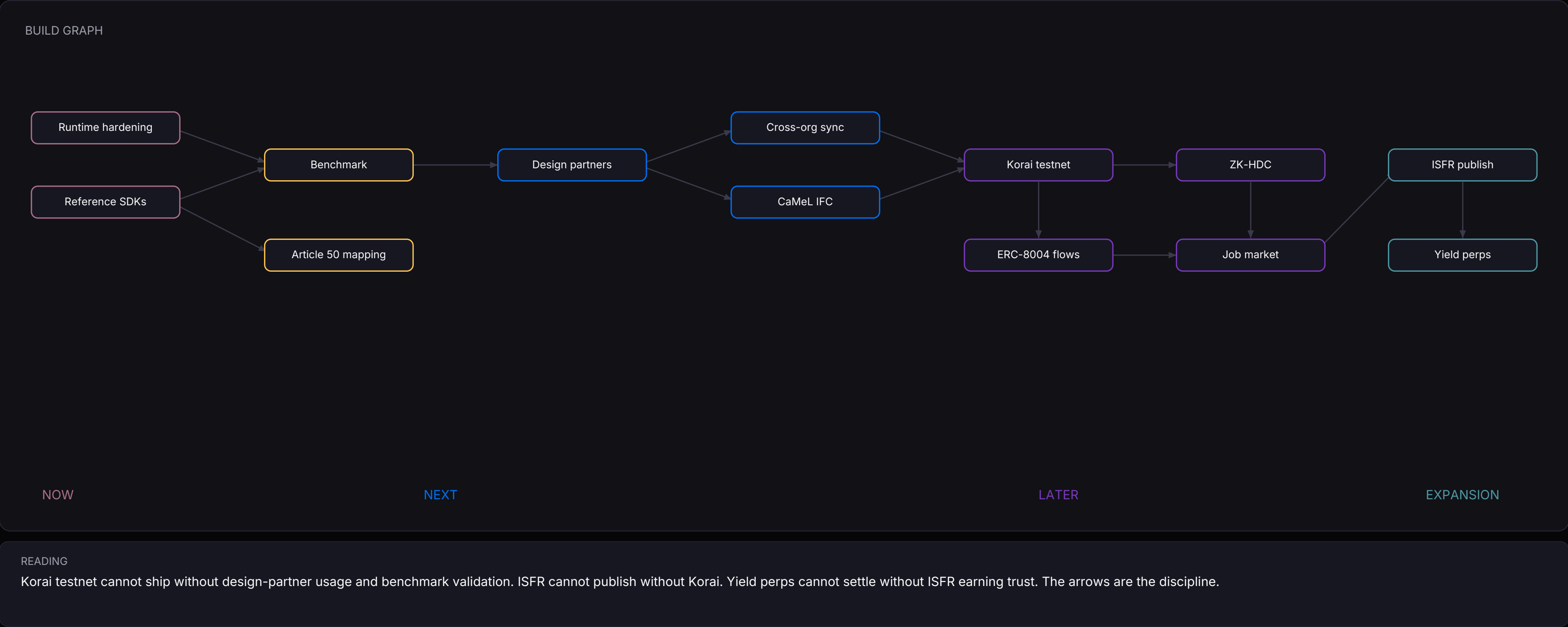
PRESERVE	AVOID
Agent Coordination Plane Structural category, not aspirational.	Trust layer Crowded and vague.
Durable runtime for production agents Clear buyer value.	Agent OS Overused and unclear.
Coordination before autonomy Avoids AGI / autonomy overclaim.	Stripe for agents Payment rails are not the category.
Compliance artifacts as byproduct Useful and legally careful.	Blockchain company Narrows audience and inverts sequencing.
Token-path economics Allows protocol economics without token-first framing.	Fully autonomous agents Creates safety and skepticism burden.
The model is the same; the system is the variable. Core thesis.	Solves EU AI Act Legal overclaim.
The model is commoditizing. The scaffold is the moat. Durable moat framing.	Guaranteed 30× savings Benchmark overclaim without methodology.
Models execute. Nunchi coordinates. Short version.	79% coordination failures Not supported by fetched MAST page.

SOURCES

1. MAST — Multi-Agent Failure Taxonomy — <https://arxiv.org/abs/2503.13657>

What blocks what.

The build sequence is a graph, not a list. Every later milestone has a hard dependency on earlier proof.



Each major claim mapped to its strongest external source.

If a claim does not appear here, treat it as internal synthesis — design target or speculative.

AREA	CLAIM	SOURCE	LINK
PROBLEM	Multi-agent failure taxonomy	MAST	https://arxiv.org/abs/2503.13657
PROBLEM	Cost under-weighted in evaluation	AI Agents That Matter	https://arxiv.org/abs/2407.01502
SECURITY	Architecture > detection	CaMeL	https://arxiv.org/abs/2503.18813
SECURITY	Adaptive attacks beat 8 defenses	Adaptive Prompt Injection	https://arxiv.org/abs/2503.00061
COST	Prompt caching at 0.1× input price	Anthropic prompt-caching docs	https://docs.anthropic.com/en/docs/build-with-claude/prompt-caching
COST	Up to 90% cost / 85% latency	AWS Bedrock prompt caching	https://aws.amazon.com/blogs/aws/reduce-costs-and-latency-with-amazon-bedrock-intelligent-prompt-routing-and-prompt-caching-preview/
COST	RouteLLM > 2× cost reduction	RouteLLM	https://arxiv.org/abs/2406.18665
COST	FrugalGPT 98% cost reduction	FrugalGPT	https://arxiv.org/abs/2305.05176
PROTOCOL	MCP 97M / mo SDK	MCP joins AAIF	https://blog.modelcontextprotocol.io/posts/2025-12-09-mcp-joins-agentic-ai-foundation/
PROTOCOL	A2A 150+ orgs, 5 SDKs	Linux Foundation A2A	https://www.linuxfoundation.org/press/a2a-protocol-surpasses-150-organizations-lands-in-major-cloud-platforms-and-sees-enterprise-producti...
PROTOCOL	ERC-8004 registries	ERC-8004	https://eips.ethereum.org/EIPS/eip-8004
PROTOCOL	x402 HTTP payments	Coinbase x402	https://docs.cdp.coinbase.com/x402/welcome
REG	Article 50 transparency	EU AI Act, Article 50	https://artificialintelligenceact.eu/article/50/
REG	Article 99 penalties	EU AI Act, Article 99	https://artificialintelligenceact.eu/article/99/
REG	Article 113 timing	EU AI Act, Article 113	https://artificialintelligenceact.eu/article/113/
MARKET	OTC derivatives notional	BIS OTC stats	https://www.bis.org/publ/otc_hy2512.htm
MARKET	OTC IRD turnover	BIS IRD turnover	https://www.bis.org/statistics/rpfx25_ir.htm
MEMORY	PathHD KG retrieval	PathHD	https://www.semanticscholar.org/paper/9ccbef593ba1e4eb5009b7477493d687f72bb9ed
MEMORY	Bit-Level binary HDC retrieval	Bit-Level Semantics	https://ieeexplore.ieee.org/document/11282933/

RULE

Every external reference appearing in this whitepaper is consolidated on the citation source map (next page). If a number is not in this map, it should be labeled ‘derived’ or ‘internal’ before publication.

Every URL used in this document.

Hyperlinks are clickable. No URL appears anywhere in this document that is not present in the source markdown corpus.

KEY	TITLE	URL
MAST	Multi-Agent Failure Taxonomy	https://arxiv.org/abs/2503.13657
AIAM	AI Agents That Matter	https://arxiv.org/abs/2407.01502
ANTHROPIC_CACHE	Claude prompt caching documentation	https://docs.anthropic.com/en/docs/build-with-claude/prompt-caching
AWS_BEDROCK	AWS Bedrock prompt caching and routing	https://aws.amazon.com/blogs/aws/reduce-costs-and-latency-with-amazon-bedrock-intelligent-prompt-routing-and-prompt-caching-preview/
ROUTE_LLM	RouteLLM	https://arxiv.org/abs/2406.18665
FRUGALGPT	FrugalGPT	https://arxiv.org/abs/2305.05176
CAMEL	CaMeL — capability-based agent security	https://arxiv.org/abs/2503.18813
ADAPTIVE	Adaptive Prompt Injection Attacks	https://arxiv.org/abs/2503.00061
MCP	MCP joins Agentic AI Foundation	https://blog.modelcontextprotocol.io/posts/2025-12-09-mcp-joins-agentic-ai-foundation/
A2A	Linux Foundation A2A announcement	https://www.linuxfoundation.org/press/a2a-protocol-surpasses-150-organizations-lands-in-major-cloud-platforms-and-sees-enterprise-production-use-in-first-year
ERC8004	ERC-8004 — Trustless Agents	https://eips.ethereum.org/EIPS/eip-8004
X402	Coinbase x402 documentation	https://docs.cdp.coinbase.com/x402/welcome
EU_50	EU AI Act, Article 50	https://artificialintelligenceact.eu/article/50/
EU_99	EU AI Act, Article 99	https://artificialintelligenceact.eu/article/99/
EU_113	EU AI Act, Article 113	https://artificialintelligenceact.eu/article/113/
BIS_OTC	BIS OTC derivatives statistics, Jun-2025	https://www.bis.org/publ/otc_hy2512.htm
BIS_IRD	BIS OTC IRD turnover, April 2025	https://www.bis.org/statistics/rpfx25_ir.htm
PATHHD	PathHD — encoder-free HDC for KG reasoning	https://www.semanticscholar.org/paper/9ccbef593ba1e4eb5009b7477493d687f72bb9ed
BITLEVEL	Bit-Level Semantics — HDC retrieval	https://ieeexplore.ieee.org/document/11282933/

What was retired. What was preserved.

The rewrite tightened claims and reframed sequencing. The thesis is unchanged.

RETIRED / SOFTENED	PRESERVED
'79% of multi-agent failures are coordination' Removed — fetched MAST page does not state this.	'The scaffold is the product' Main thesis and conclusion.
'30× cost reduction' as a universal claim Softened to 10–20× target; 30× only with explicit methodology.	'The model is the same. The system is the variable.' Executive thesis, roadmap, conclusion.
'Trust layer for agents' Retired in favor of Agent Coordination Plane.	Roko + Korai two-part system Product architecture.
'Blockchain/company-first' framing Reordered: runtime first, chain after proof.	Roko runtime loop Product architecture and appendix.
'ISFR as beachhead' Reframed as expansion domain.	Signal / Pulse / Cell Architecture sections.
'Article 50 compliance solved' Reframed as provenance/audit artifacts.	Predict-publish-correct Appendix learning loop.
'HDC as already commercially proven' Reframed as strong technical bet, thin commercial traction.	Gate pipeline as enterprise control Whitepaper and appendix.
'Token-first economics' Reframed as deferred protocol economics.	HDC/ZK-HDC as proof-friendly memory Whitepaper and appendix with caveats.
	Korai as sovereign coordination chain Whitepaper and appendix.
	Token deferral Business model and risk sections.

SOURCES

1. MAST — Multi-Agent Failure Taxonomy — <https://arxiv.org/abs/2503.13657>

Who is buying — and what they hear.

Four personas across the typical enterprise org chart. Each has a distinct pain that the coordination plane addresses.

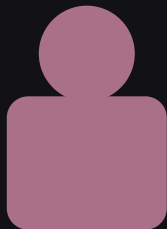


AI Platform Lead

PAIN
Agent prototypes are expensive, brittle, hard to operate.

ANGLE
Lower cost, replay, routing, gates, runtime governance.

METRIC
Cost per task · time-to-replay · gate pass-rate



AI Governance / Risk Lead

PAIN
Agents cannot move to production without identity, provenance, audit.

ANGLE
Compliance artifacts as byproduct of coordination.

METRIC
Article 50 mapping coverage · audit-trace completeness



Engineering Leader

PAIN
Frameworks don't solve cross-agent memory, cost, and policy.

ANGLE
Durable runtime and control plane underneath frameworks.

METRIC
MTTR · cross-agent memory hits · cost per resolution



Design Partner CTO

PAIN
Differentiated agent infra is needed without building it in-house.

ANGLE
Forward-deployed support + runtime extensibility.

METRIC
Time-to-first-workload · co-built features

Resolve before investor distribution.

These do not block the rewrite — but they should be resolved before public distribution.

Korai naming

Should ‘Korai’ replace ‘Nunchi chain’ in market-facing materials?

The rewrite standardizes on Korai — but naming should be locked before publication.

Roko license

Is Roko Apache 2.0 already public, or still private/internal?

Public wording depends on actual repository status.

nunchi run --share

Is the shareable replay artifact implemented?

Treated as a critical not-yet-built artifact based on the source corpus.

Demo benchmark

Which canonical task set, methodology, and number for the demo?

Source corpus contains conflicting 30× examples; pick one and stick to it.

ISFR placement

Stay in main whitepaper, or move fully to appendix for non-crypto investors?

Currently kept as expansion section after the coordination thesis.

Audience ladder

Public whitepaper, investor memo, internal strategy, technical litepaper — different versions?

Each requires a different level of disclosure (see story split page).

Every Nunchi document passes this checklist.

If a doc fails any row, do not ship it.

CHECKLIST	
✓ Category Uses Agent Coordination Plane.	✓ Product hierarchy Roko first, Korai second, ISFR later.
✓ Claims Distinguishes current, internal, externally supported, roadmap.	✓ Cost Does not state 30× without methodology.
✓ Regulation Does not claim legal compliance as a standalone outcome.	✓ Chain Does not imply production chain is live unless it is.
✓ Token Does not lead with token mechanics.	✓ Moat Centers memory, calibration, reputation, settlement, scaffold compounding.
✓ Tone Sells inevitability through architecture, not hype.	✓ Sources Every external claim has a clickable URL footnote.

Avoid ‘traction’ language until external traction exists.

Use this ladder to describe progress. Each rung is unlocked by the rung below.



One picture, the whole thesis.

Read top-to-bottom: problem → category → product → wedge → moat → expansion.

PROBLEM	Coordination gap between prototype and production	bottleneck
CATEGORY	Agent Coordination Plane (separates plane from execution)	frame
PRODUCT	Roko (runtime) · Korai (chain) · primitives (Signal/Pulse/Cell)	scaffold
WEDGE	Cost: cache × route × gate (10–20× target with methodology)	wedge
MOAT	Scaffold: knowledge · calibration · reputation · settlement	compound
EXPANSION	ISFR — first financial primitive on the coordination network	primitive

REMEMBER

The model is commoditizing. The scaffold is the moat.
Models execute. Nunchi coordinates.

END OF DOCUMENT

Models execute. Nunchi coordinates.

The model is the same. The system is the variable.
The model is commoditizing. The scaffold is the moat.

NEXT STEPS

- Read**
Strategy whitepaper · diligence appendix · source-of-truth index
- Run**
nunchi run agent.toml --task R15 — see demo storyboard
- Share**
nunchi run --share for replayable inspection
- Audit**
nunchi audit for compliance and provenance artifacts